

Exponential equations:
Exponents and Logarithms

$$u = \log_a(x) \Rightarrow \left| \begin{array}{l} x = a^u \end{array} \right.$$

$$v = \log_a(y) \Rightarrow \left| \begin{array}{l} y = a^v \end{array} \right.$$

$$xy = a^u \cdot a^v$$

$$xy = a^{u+v}$$

$$xy = a^{\log_a(x) + \log_a(y)}$$

$$\log_a(xy) = \log_a(x) + \log_a(y)$$

This is the complete proof of the property of multiplying logarithms, without any memorization of laws, only returning to the definition.

E-Vex

$$\log_a(x^r) = r \cdot \log_a(x)$$

$$\Rightarrow u = \log_a(x)$$

$$r \cdot u = r \cdot \log_a(x)$$

$$\Rightarrow x^r = a^{r \cdot u}$$

$$\log_a(x^r) = r \cdot u$$

$$\log_a(x^r) = r \cdot \log_a(x)$$

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